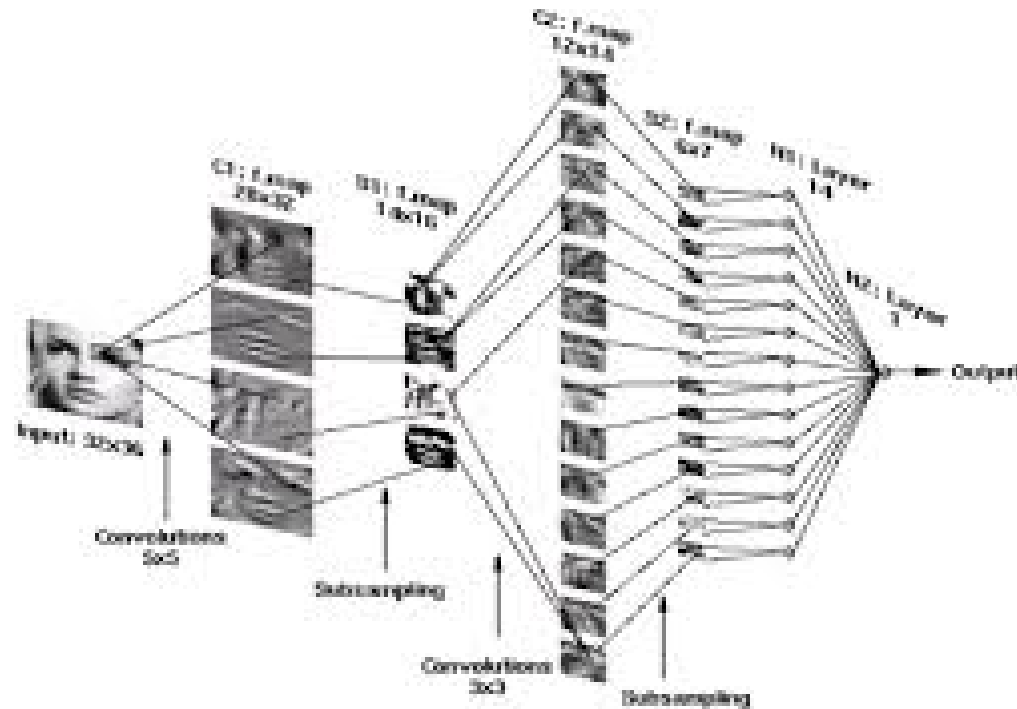


Mathematics of Machine Learning

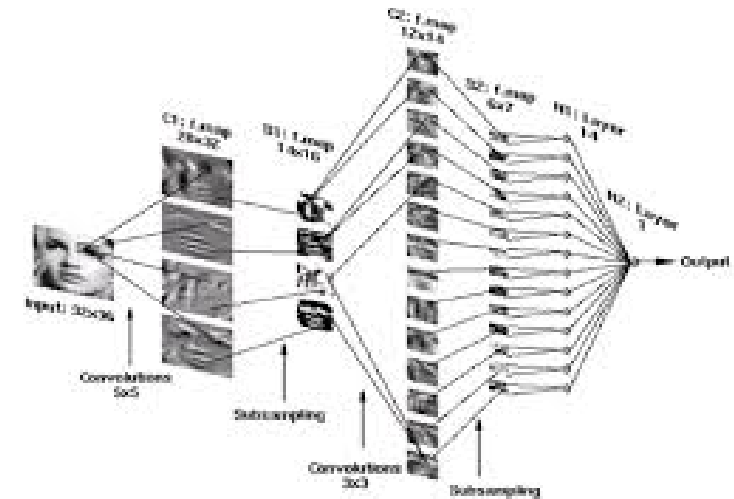
Introduction



Course Details and Topics

Outline

1. **Why are you doing this course?**
2. What we assume that you know



Orientation

- This course addresses three cohorts of students:
 - ★ Third year Artificial Intelligence Students
 - ★ MSc in Machine Learning Students
 - ★ Volunteers
- It is dual coded (AICE3001/AICE6002), with subtle difference in expectations reflected in slightly different evaluation
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- LLMs can help us a lot, but if you don't trust yourself you are disempowered
- This course is aimed at building up trust in your understanding of how machine learning works

Why Mathematics?

- Learning machines are notoriously difficult to understand
- Mathematics allows us to reason rigorously
- Posing question mathematically and seeking mathematical solutions provide allow us to build a strong intuition
- Mathematics also allow us to describe the world precisely
- It also provides an arsenal of tools that turn out to be hugely useful

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Cracking the Code

- Mathematics is a language that takes time to master
- For some it is easy, for others it will be a struggle
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Starting from Different Places

- You all start from different places
- Some of you may have a degree in mathematics
- For others it may have been your least favourite subject
- I invite strong mathematicians to become even stronger
- For students starting from a weak base, judge yourselves by your progress, not by comparing yourself to others

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The Journey

- We are going to visit lots of aspects of mathematics that have proved fruitful in machine learning
- Some of this will be very concrete, while in other cases by look at a higher level of abstraction we can generalise methods across different application areas
- We will cover, calculus, vector spaces, convexity, optimisation, entropy, probability, statistics and much else besides
- The journey will pay you off, these are the foundations of most of machine learning and will give you super-powers that are hard to acquire without mathematics

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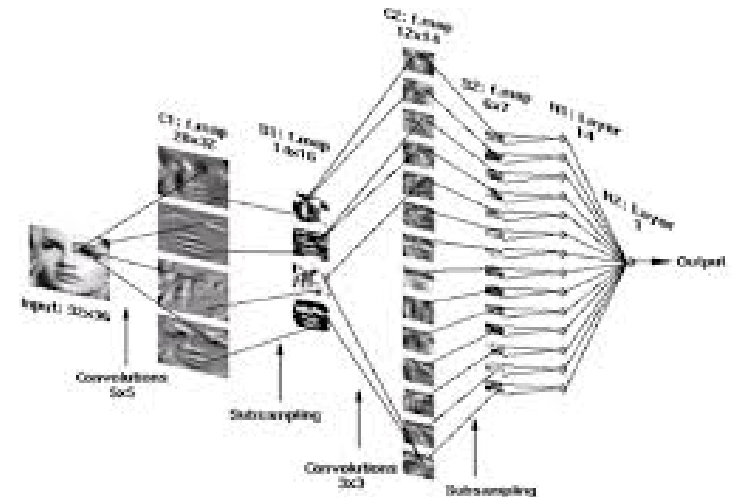
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What we Assume you Know

- You should all have A-level mathematics and have done some mathematics at Uni
- You should know what a function is, e.g. $f(x) = x^2 + 3$
- You should be able to do simple algebraic manipulations
- And some elementary calculus

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Vectors

- Ordinary vectors (what mathematicians call elements of $\mathbb{R}^n$) are much used in machine learning and should be familiar
- We will use bold symbols to represent **column vectors**, e.g.

$$\boldsymbol{x} = \begin{pmatrix} 1.2 \\ 2.3 \\ 4.7 \end{pmatrix}$$

- We denote **row vectors** as the **transpose** of column vectors
 $\boldsymbol{x}^T = (1.2, 2.3, 4.7)$
- We care whether vectors are row or column vectors as it allows us to write equations involving vectors succinctly and unambiguously

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Matrices

- Matrices are blocks of number that define linear mappings between vectors, e.g. $\mathbf{M} \in \mathbb{R}^{2 \times 3}$

$$\mathbf{M} = \begin{pmatrix} 1.1 & -2.1 & 6.2 \\ 9.3 & 3.1 & -4.4 \end{pmatrix}$$

- An $m \times n$ matrix (i.e. $\mathbf{M} \in \mathbb{R}^{m \times n}$) has m rows and n columns
- An $m \times n$ matrix acting on a vector in $\mathbb{R}^n$ transforms it to $\mathbb{R}^m$ vector, e.g.

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Composition of Mappings

- Matrices, $\mathbf{A}$, $\mathbf{B}$, etc. represent mappings between spaces
- Provided the matrices are the correct size we can compose these mappings, e.g. if $\mathbf{A} \in \mathbb{R}^{\ell \times m}$ and $\mathbf{B} \in \mathbb{R}^{m \times n}$ then $\mathbf{AB}$ represents the effect of
 - ★ first applying $\mathbf{B}$ to vectors in $\mathbb{R}^n$ to obtain vectors in $\mathbb{R}^m$
 - ★ and then applying $\mathbf{A}$ to get vectors in $\mathbb{R}^\ell$

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Matrix Multiplication

- The combined mapping $\mathbf{AB}$ can be represented by an $\ell \times n$ matrix $\mathbf{C}$ which we can compute using matrix multiplication (rows times columns)

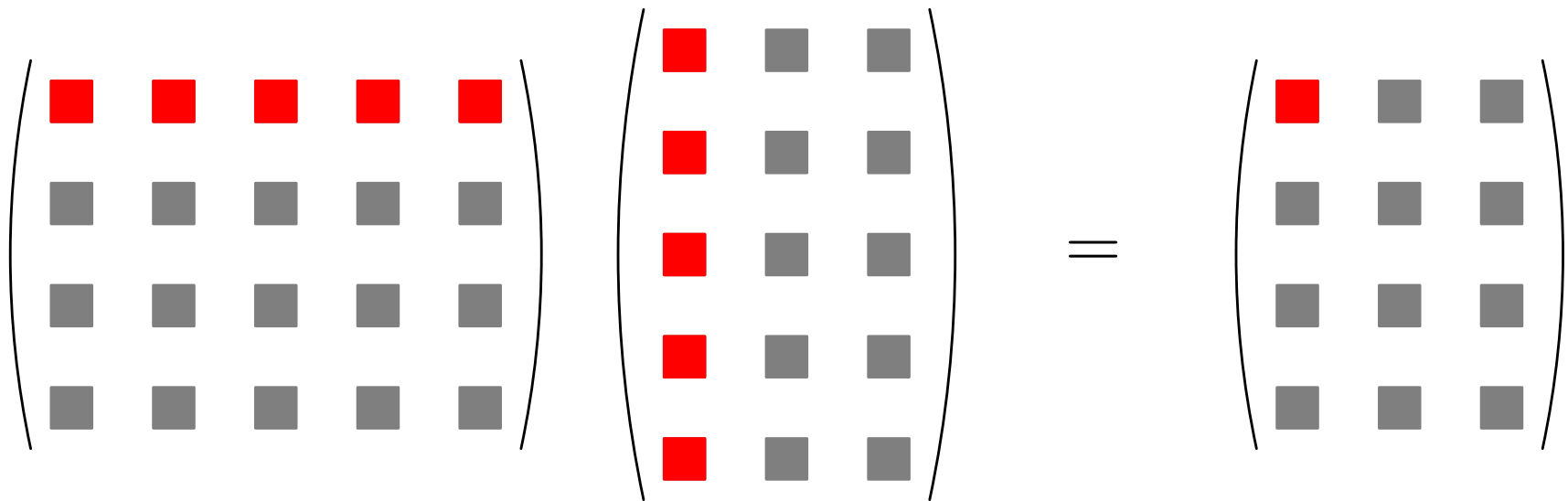
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The diagram illustrates matrix multiplication using a visual representation of matrices as grids of squares. The first matrix, representing $\mathbf{A}$, is a 5x5 grid with 5 rows and 5 columns. The second matrix, representing $\mathbf{B}$, is a 5x3 grid with 5 rows and 3 columns. An equals sign follows, leading to the resulting matrix $\mathbf{C}$, which is a 5x3 grid with 5 rows and 3 columns. This visualizes the operation $\mathbf{AB} = \mathbf{C}$.

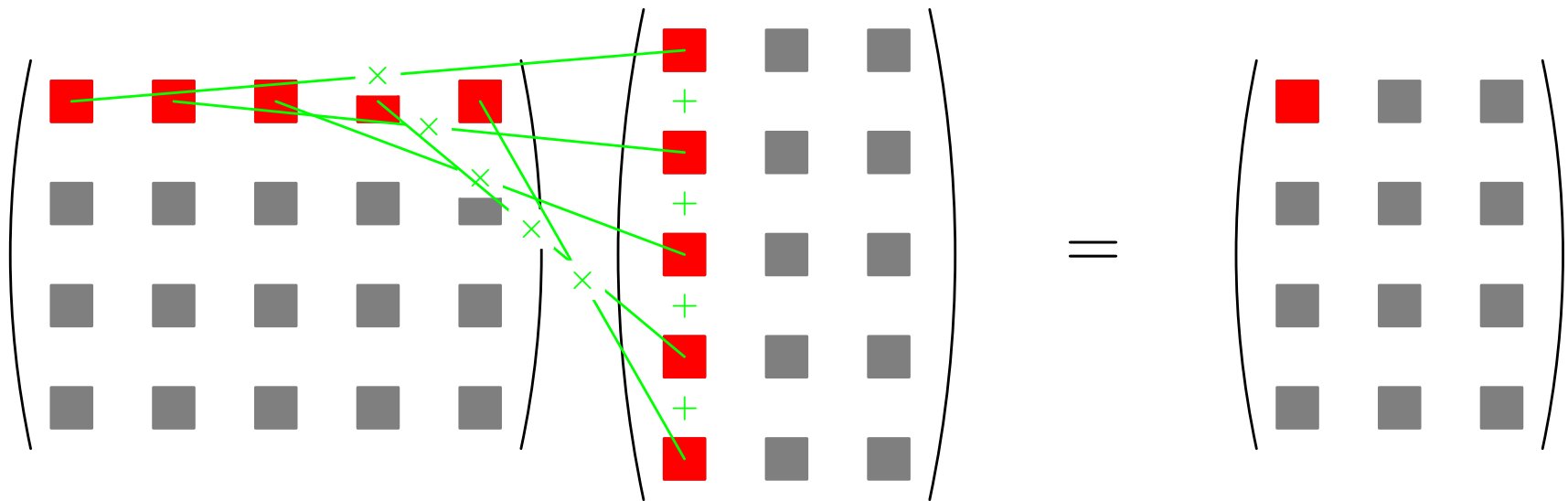
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$$\begin{pmatrix} \text{red} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \end{pmatrix} \begin{pmatrix} \text{gray} & \text{red} & \text{gray} \\ \text{gray} & \text{red} & \text{gray} \\ \text{gray} & \text{red} & \text{gray} \\ \text{gray} & \text{red} & \text{gray} \\ \text{gray} & \text{red} & \text{gray} \end{pmatrix} = \begin{pmatrix} \text{gray} & \text{red} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} \end{pmatrix}$$

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The diagram illustrates the matrix multiplication $\mathbf{AB} = \mathbf{C}$ using a grid of squares. The first matrix (4x5) has a top row of red squares and four rows of gray squares. The second matrix (5x3) has four rows of gray squares and a rightmost column of red squares. The result matrix (4x3) has four rows of gray squares and a rightmost column of red squares. The matrices are enclosed in large parentheses, and an equals sign is placed between the second and third matrices.

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The diagram illustrates matrix multiplication using a grid of squares. The first matrix is 4 rows by 5 columns. The second matrix is 5 rows by 3 columns. The result is a 4 rows by 3 columns matrix. The red squares indicate the specific row and column being multiplied to produce a single element in the result matrix.

Gray	Gray	Gray	Gray	Gray	=	Gray	Gray	Gray
Red	Red	Red	Red	Red		Gray	Red	Gray
Gray	Gray	Gray	Gray	Gray		Gray	Gray	Gray
Gray	Gray	Gray	Gray	Gray		Gray	Gray	Gray
Gray	Gray	Gray	Gray	Gray		Gray	Gray	Gray

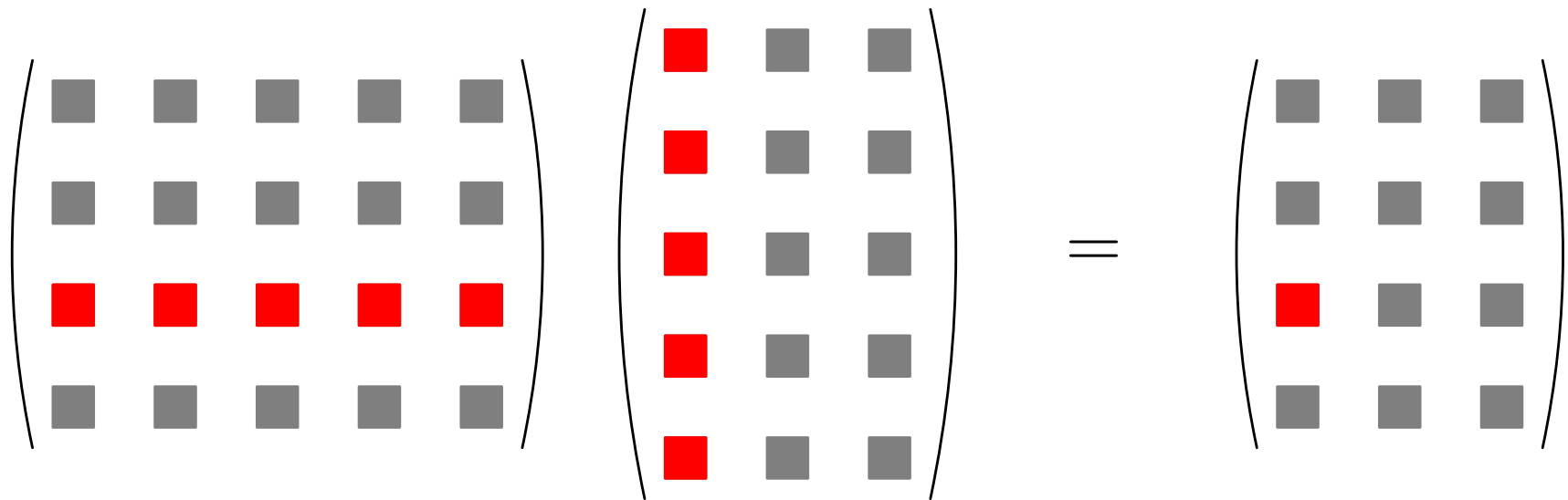
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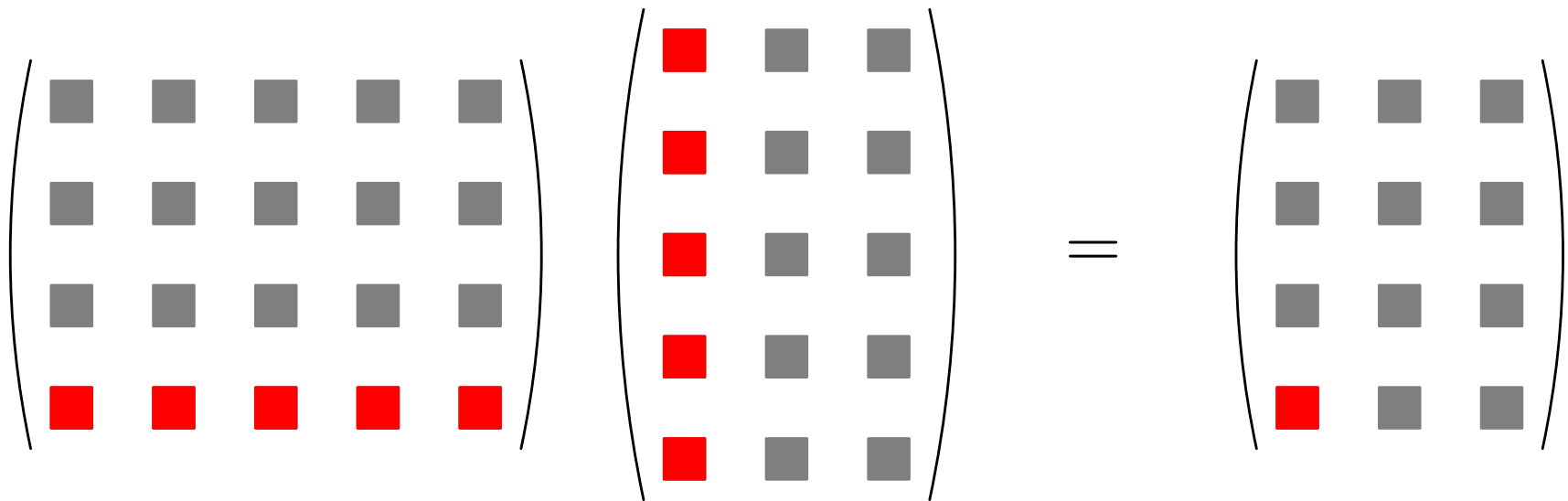
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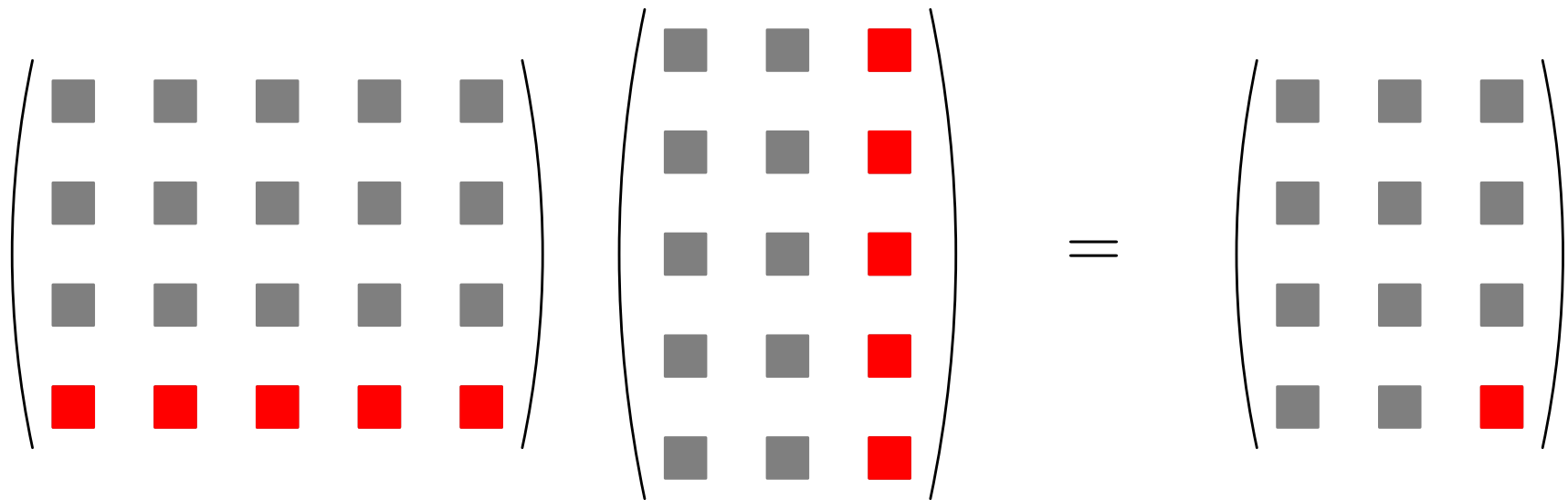
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The diagram illustrates matrix multiplication using a grid of squares. The first matrix is a 4x5 grid with 4 rows and 5 columns. The bottom row is highlighted in red, while the other three rows are gray. The second matrix is a 5x3 grid with 5 rows and 3 columns. The middle column is highlighted in red, while the other two columns are gray. An equals sign follows, leading to the resulting 4x3 matrix. This matrix has 4 rows and 3 columns. The bottom-middle square (row 4, column 2) is highlighted in red, while the other squares are gray.

Matrix Multiplication

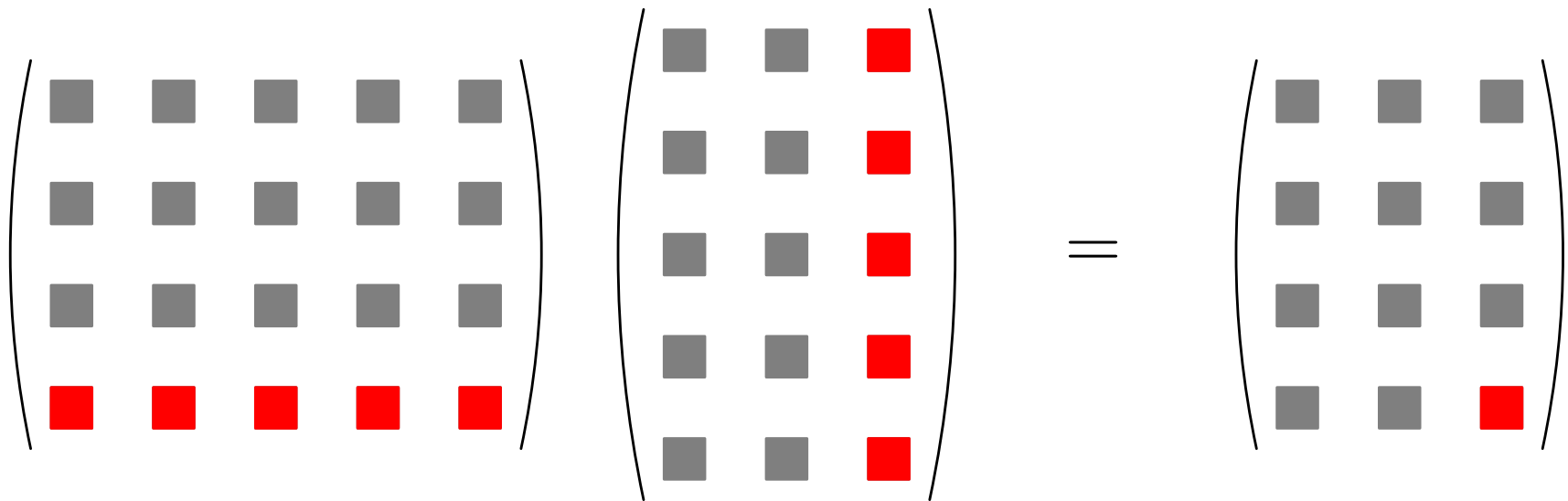
- The combined mapping $\mathbf{AB}$ can be represented by an $\ell \times n$ matrix $\mathbf{C}$ which we can compute using matrix multiplication (rows times columns)



The diagram illustrates matrix multiplication using a grid of squares. The first matrix is a 4x5 grid with 4 rows and 5 columns. The first three rows consist of 5 gray squares each. The fourth row consists of 5 red squares. The second matrix is a 5x3 grid with 5 rows and 3 columns. The first four rows consist of 2 gray squares followed by 1 red square. The fifth row consists of 2 gray squares followed by 1 red square. An equals sign follows, leading to the resulting 4x3 matrix. The first three rows consist of 3 gray squares each. The fourth row consists of 2 gray squares followed by 1 red square.

Matrix Multiplication

- The combined mapping $\mathbf{AB}$ can be represented by an $\ell \times n$ matrix $\mathbf{C}$ which we can compute using matrix multiplication (rows times columns)



- Matrix multiplication is associative: $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$

Matrix Multiplication

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The diagram illustrates matrix multiplication using a grid of squares. The first matrix is 4 rows by 5 columns, with 4 gray squares in each row and 1 red square in each row (bottom row). The second matrix is 5 rows by 3 columns, with 4 gray squares in each row and 1 red square in each row (rightmost column). The result is a 4 rows by 3 columns matrix, with 3 gray squares in each row and 1 red square in each row (rightmost column). The matrices are enclosed in large parentheses, and an equals sign is placed between the second and third matrices.

- Matrix multiplication is associative: $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$
- But in general not commutative $\mathbf{AB} \neq \mathbf{BA}$

Transpose

- We often encounter transposes of a matrix $\mathbf{M}^T$ where we have exchanged the rows and columns: $[\mathbf{M}^T]_{ij} = [\mathbf{M}]_{ji}$
- Clear $(\mathbf{M}^T)^T = \mathbf{M}$
- It is simple question of book keeping to show that

$$(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$$

- For a symmetric matrix $\mathbf{M} = \mathbf{M}^T$
- Since 1×1 matrices are always symmetric

$$\mathbf{a}^T \mathbf{b} = (\mathbf{a}^T \mathbf{b})^T = \mathbf{b}^T \mathbf{a}$$

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$$\begin{aligned} \mathbf{a}^\top \mathbf{b} &= (\mathbf{a}^\top \mathbf{b})^\top = \mathbf{b}^\top \mathbf{a} \\ \mathbf{a}^\top \mathbf{M} \mathbf{b} &= \mathbf{b}^\top \mathbf{M}^\top \mathbf{a} \end{aligned}$$

Matrix-Vector Multiplication

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The diagram illustrates matrix-vector multiplication as matrix multiplication. On the left, a 4x5 matrix is represented by a grid of 20 gray squares enclosed in large parentheses. To its right is a column vector, represented by a single column of 4 gray squares enclosed in large parentheses. An equals sign follows, and to the right is the resulting column vector, represented by a single column of 4 gray squares enclosed in large parentheses. This visualizes the operation $(4 \times 5) \times (5 \times 1) = (4 \times 1)$.

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The diagram illustrates matrix-vector multiplication as matrix multiplication. It shows a 4x5 matrix (with 5 red squares in the top row and 4 gray squares in each of the other three rows) multiplied by a 5x1 column vector (with 5 red squares). This is followed by an equals sign and a 4x1 column vector (with 1 red square in the top row and 3 gray squares in the other rows), representing the resulting vector.

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The diagram illustrates matrix-vector multiplication as matrix multiplication. On the left, a 5x5 matrix is shown with its second row highlighted in red. This matrix is multiplied by a 5x1 column vector, also with its second element highlighted in red. The result is a 5x1 column vector where only the second element is highlighted in red, representing the dot product of the second row of the matrix and the column vector.

Matrix-Vector Multiplication

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The diagram illustrates matrix-vector multiplication as matrix multiplication. It shows a 5x5 matrix (represented by a grid of squares, with the third row highlighted in red) multiplied by a 5x1 column vector (represented by a single column of squares, with all squares highlighted in red). The result is a 5x1 column vector (represented by a single column of squares, with the third square highlighted in red). The equation is shown as:

$$\begin{pmatrix} \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{red} & \text{red} & \text{red} & \text{red} & \text{red} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \\ \text{gray} & \text{gray} & \text{gray} & \text{gray} & \text{gray} \end{pmatrix} \begin{pmatrix} \text{red} \\ \text{red} \\ \text{red} \\ \text{red} \\ \text{red} \end{pmatrix} = \begin{pmatrix} \text{gray} \\ \text{gray} \\ \text{red} \\ \text{gray} \\ \text{gray} \end{pmatrix}$$

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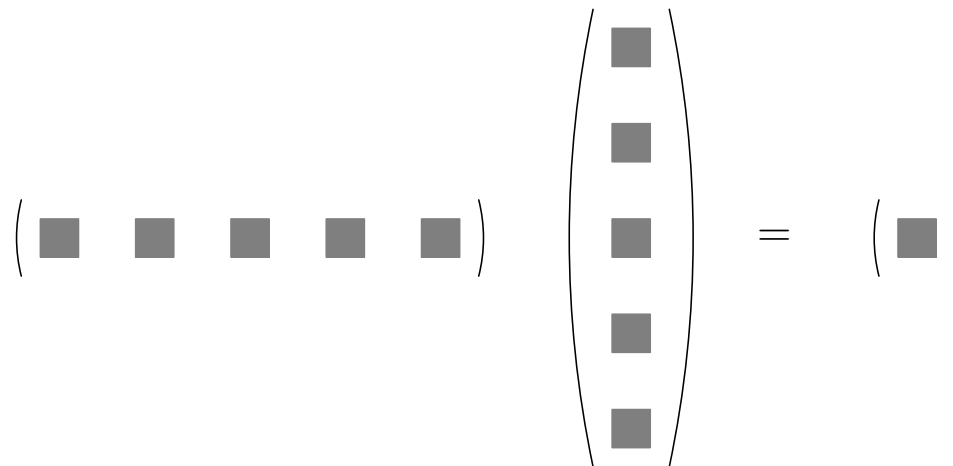
The diagram illustrates matrix-vector multiplication as matrix multiplication. On the left, a 4x5 matrix is shown with 4 rows and 5 columns. The first three rows contain gray squares, and the fourth row contains red squares. This matrix is multiplied by a 5x1 column vector on the right, which consists of five red squares. An equals sign follows, and to the right is a 4x1 column vector consisting of four gray squares, representing the result of the multiplication.

Inner Products

- The dot or inner-product between two vectors with the same number of elements, can be written $\mathbf{a}^T \mathbf{b}$

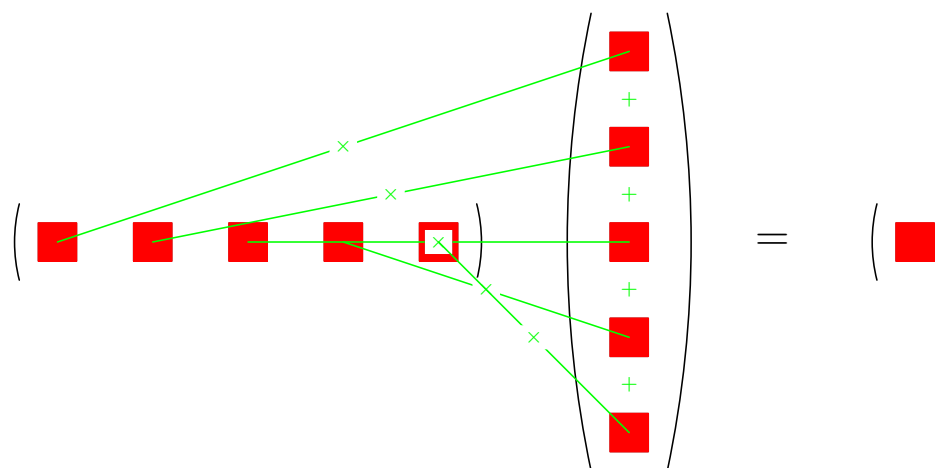
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$$\left(\begin{array}{ccccc} \blacksquare & \blacksquare & \blacksquare & \blacksquare & \blacksquare \end{array} \right) = \left(\begin{array}{c} \blacksquare \\ \blacksquare \\ \blacksquare \\ \blacksquare \\ \blacksquare \end{array} \right) = \left(\blacksquare \right)$$

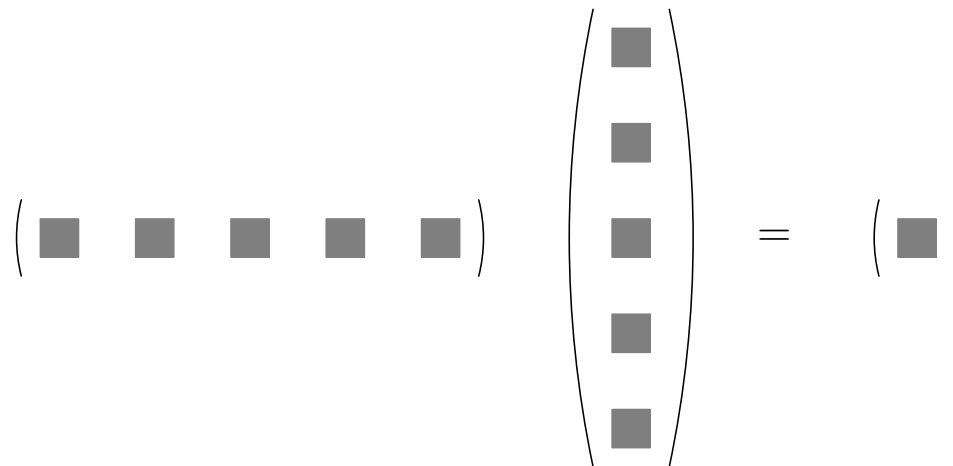
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$$\begin{pmatrix} \blacksquare & \blacksquare & \blacksquare & \blacksquare & \blacksquare \end{pmatrix} \begin{pmatrix} \blacksquare \\ \blacksquare \\ \blacksquare \\ \blacksquare \\ \blacksquare \end{pmatrix} = \begin{pmatrix} \blacksquare \end{pmatrix}$$

- Inner products encode angles between vectors

$$\mathbf{a}^T \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos(\theta)$$

Outer Products

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 ab^T

Outer Products

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The diagram illustrates the outer product operation ab^T . On the left, a column vector a of size 4x1 is represented by a large left parenthesis followed by four gray squares stacked vertically, and a large right parenthesis. This is multiplied by a row vector b^T of size 1x3, represented by a large left parenthesis followed by three gray squares, and a large right parenthesis. An equals sign follows, leading to the resulting 4x3 matrix. This matrix is shown as a large left parenthesis followed by four rows, each containing three gray squares, and a large right parenthesis.

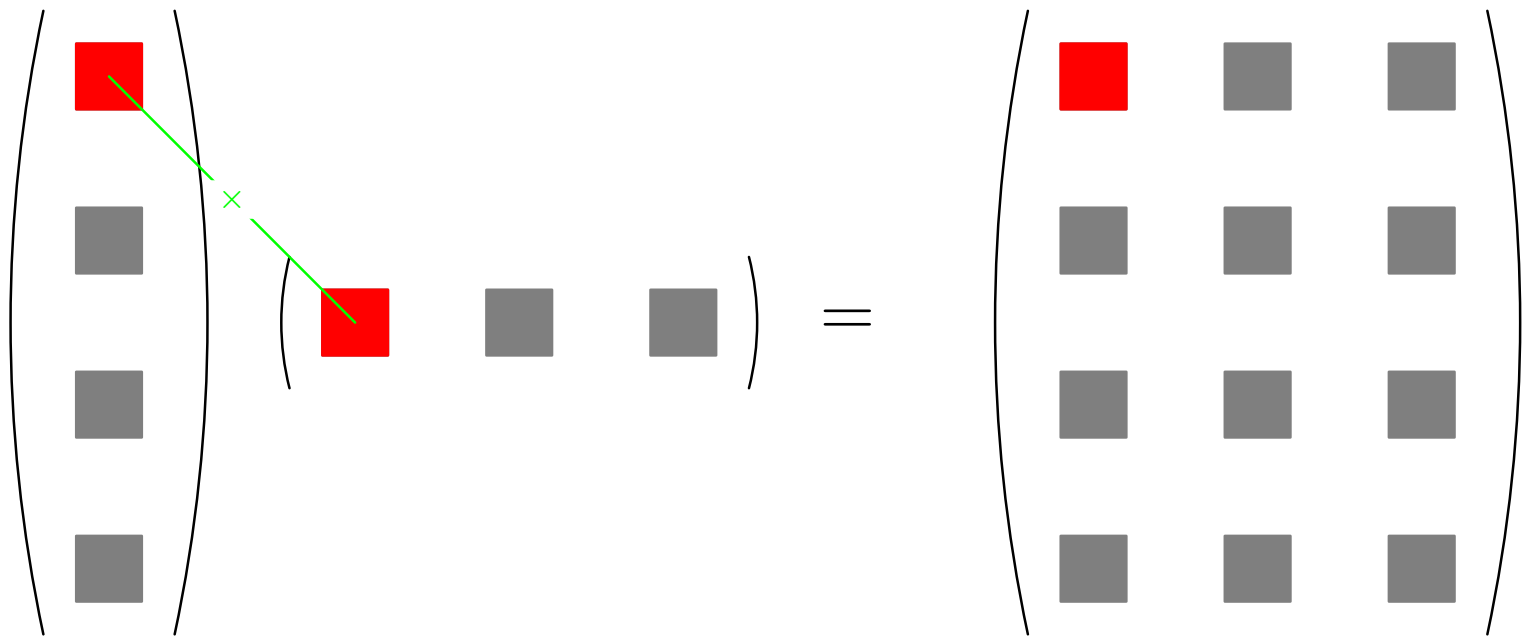
Outer Products

- We can also define the outer-product between any size vectors, ab^T

The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4 is shown, with its top element highlighted in red and the others in gray. This is multiplied by a row vector of size 3, also with its first element highlighted in red and the others in gray. An equals sign follows, leading to a 4x3 matrix on the right. Each row of this matrix corresponds to one of the four elements of the first vector. The first row contains one red square followed by two gray squares, representing the first row of the vector multiplied by the entire second vector. The subsequent three rows each contain three gray squares, representing the remaining rows of the first vector multiplied by the entire second vector.

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$$\begin{pmatrix} \text{red} \\ \text{gray} \\ \text{gray} \\ \text{gray} \end{pmatrix} \begin{pmatrix} \text{gray} & \text{red} & \text{gray} \end{pmatrix} = \begin{pmatrix} \text{red} & \text{red} & \text{red} \\ \text{gray} & \text{red} & \text{gray} \\ \text{gray} & \text{red} & \text{gray} \\ \text{gray} & \text{red} & \text{gray} \end{pmatrix}$$

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The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4 is shown, with its top element highlighted in red and the others in gray. This is multiplied by a row vector of size 3, with its rightmost element highlighted in red and the others in gray. The result is a 4x3 matrix where each row is a scaled version of the original row vector, and the top row is highlighted in red to show the result of the outer product of the red elements.

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The diagram illustrates the outer product operation ab^T . On the left, a column vector a of size 4x1 is shown, with its second element highlighted in red. This is multiplied by a row vector b^T of size 1x3, with its first element highlighted in red. The result is a 4x3 matrix where each row is a scalar multiple of the vector b^T , and the first row is identical to b^T because the red element in a is 1. The matrix has 4 rows and 3 columns, with the first row containing one red and two gray squares, and the remaining three rows containing three gray squares each.

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The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4 is shown, with its second element highlighted in red. This is multiplied by a row vector of size 3, with its second element also highlighted in red. The result is a 4x3 matrix where every element in the second row and second column is red, while all other elements are gray.

$$\begin{pmatrix} \square \\ \color{red}\square \\ \square \\ \square \end{pmatrix} \begin{pmatrix} \square & \color{red}\square & \square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \color{red}\square & \square \\ \square & \square & \square \\ \square & \square & \square \end{pmatrix}$$

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$$\begin{pmatrix} \square \\ \color{red}\square \\ \square \\ \square \end{pmatrix} \begin{pmatrix} \square & \square & \color{red}\square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \square & \color{red}\square \\ \square & \square & \square \\ \square & \square & \square \end{pmatrix}$$

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Outer Products

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The diagram illustrates the outer product of two 4x1 vectors. On the left, a column vector of four gray squares is multiplied by a row vector of three gray squares. The result is a 4x4 matrix where the third row and third column are highlighted in red, representing the outer product of the third elements of the input vectors.

$$\begin{pmatrix} \square \\ \square \\ \square \\ \square \end{pmatrix} \begin{pmatrix} \square & \square & \square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \square & \square \\ \square & \square & \square \\ \square & \square & \square \end{pmatrix}$$

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The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4 is shown, with its third element highlighted in red. This is multiplied by a row vector of size 3, with its third element also highlighted in red. The result is a 4x3 matrix where every element in the third row and third column is red, while all other elements are gray.

$$\begin{pmatrix} \square \\ \square \\ \color{red}\square \\ \square \end{pmatrix} \begin{pmatrix} \square & \square & \color{red}\square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \square & \square \\ \color{red}\square & \color{red}\square & \color{red}\square \\ \square & \square & \square \end{pmatrix}$$

Outer Products

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The diagram illustrates the outer product operation ab^T using square blocks to represent matrix elements. On the left, a column vector a of size 4x1 is shown, with its bottom element highlighted in red. This is multiplied by a row vector b^T of size 1x3, with its first element highlighted in red. The result is a 4x3 matrix on the right, where each row is a scaled version of the row vector b^T . The first row of the resulting matrix has its first element highlighted in red, corresponding to the product of the red elements in the input vectors.

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The diagram illustrates the outer product of two vectors. On the left, a column vector of size 4 is shown, with its bottom element highlighted in red. This is multiplied by a row vector of size 3, with its middle element highlighted in red. The result is a 4x3 matrix where the bottom row of the first column is highlighted in red, representing the outer product of the red elements from the input vectors.

$$\begin{pmatrix} \square \\ \square \\ \square \\ \color{red}\square \end{pmatrix} \begin{pmatrix} \square & \color{red}\square & \square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \square & \square \\ \square & \square & \square \\ \square & \color{red}\square & \square \end{pmatrix}$$

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The diagram illustrates the outer product of a column vector and a row vector. On the left, a column vector is shown with four elements: three gray squares and one red square at the bottom. This is multiplied by a row vector with three elements: two gray squares and one red square on the right. The result is a 4x3 matrix on the right, where each element is the product of the corresponding elements from the input vectors. The bottom-right element is red, while all other elements are gray.

$$\begin{pmatrix} \square \\ \square \\ \square \\ \color{red}\square \end{pmatrix} \begin{pmatrix} \square & \square & \color{red}\square \end{pmatrix} = \begin{pmatrix} \square & \square & \square \\ \square & \square & \square \\ \square & \square & \square \\ \square & \square & \color{red}\square \end{pmatrix}$$

- Notice that we need to know whether we are dealing with row or column vectors

Norms

- Norms are non-negative numbers associated with the “size” of a vector
- The norm you should be familiar with is the **Euclidean Norm** or 2-norm

$$\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$$

- We will see there exist other norms (using different ways of measuring size)
- The Euclidean norm is intimately related to the inner-product, we will see other examples of this for different vector spaces

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Matrices as Linear Mappings

- Matrices represent mappings between vectors

$$\mathbf{b} = \mathbf{M}\mathbf{a}$$

- The elements of $\mathbf{b}$ are **linear sums** of the elements of $\mathbf{a}$

$$b_i = \sum_j M_{ij} a_j$$

- Two important consequences of this linearity are for any real c

$$\mathbf{M}(c\mathbf{a}) = c(\mathbf{M}\mathbf{a})$$

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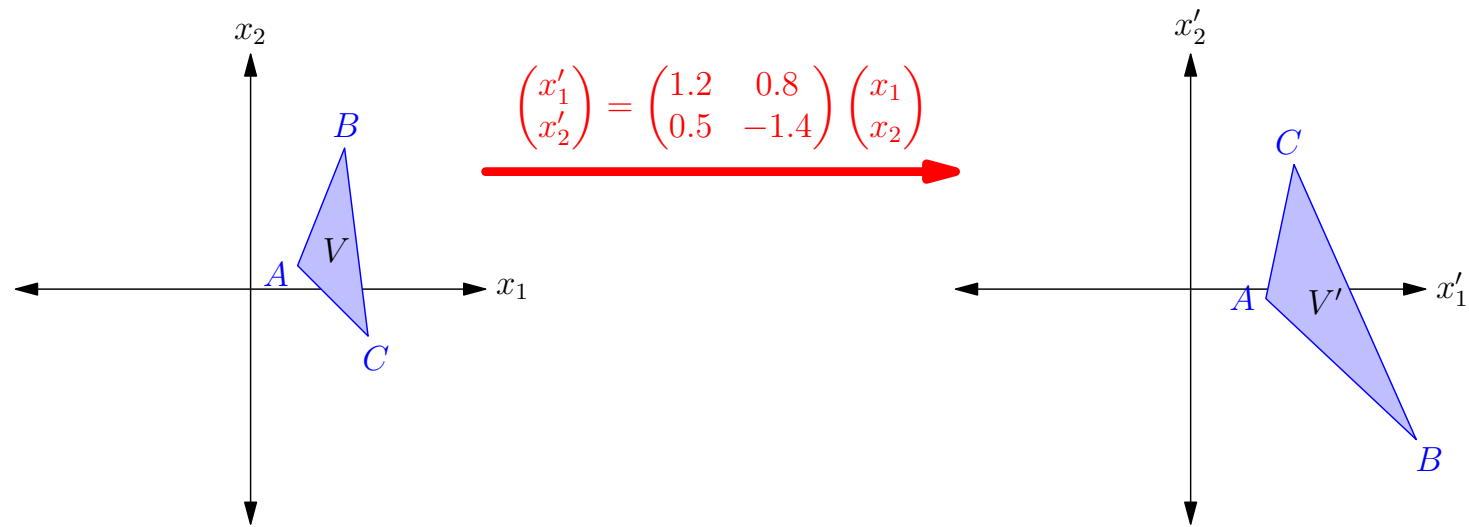
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$$\mathbf{M}(c\mathbf{a}) = c(\mathbf{M}\mathbf{a})$$

$$\mathbf{M}(\mathbf{a} + \mathbf{b}) = (\mathbf{M}\mathbf{a}) + (\mathbf{M}\mathbf{b})$$

Action Of Matrices on Volumes

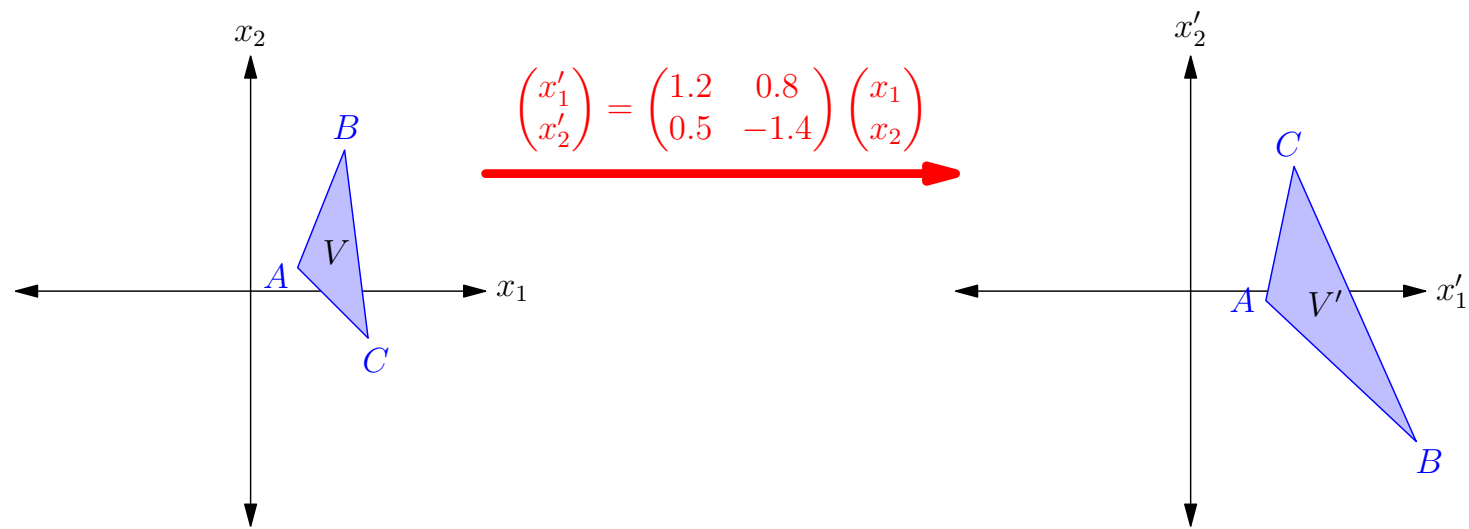
- For a closed set of points corresponding to a volume, V in $\mathbb{R}^n$, treating each point as a vector, we can ask what happens to the volume after mapping the vector by a matrix



- For square matrices, $\mathbf{M}$, the determinant is equal to $\det(\mathbf{M}) = \pm V'/V$
- The sign tells us if there has been a reflection

Action Of Matrices on Volumes

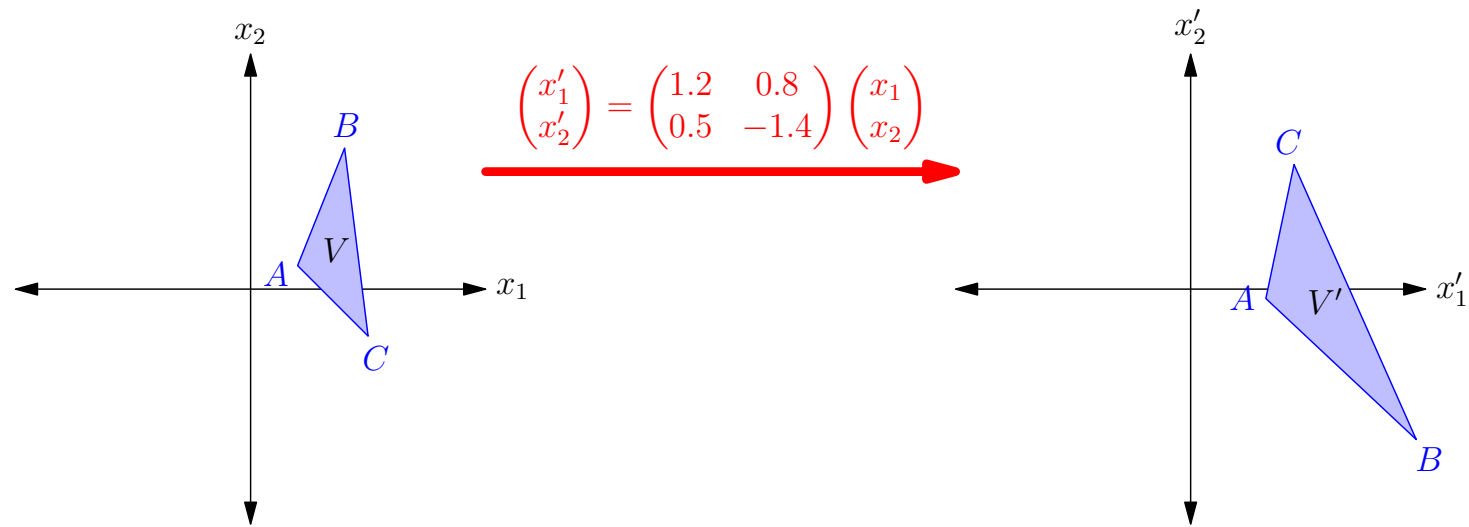
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Action Of Matrices on Volumes

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Determinants

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Multivariate Functions

- In machine learning we heavily use multivariate functions

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

- Often our functions will be scalar value $f : \mathbb{R}^n \rightarrow \mathbb{R}$
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- We say a function f is linear if

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Gradients

- The gradient of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ at a point $\mathbf{x}^*$ is defined as the column vector

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f(\mathbf{x}^*)}{\partial x_1} \\ \frac{\partial f(\mathbf{x}^*)}{\partial x_2} \\ \vdots \\ \frac{\partial f(\mathbf{x})^*}{\partial x_n} \end{pmatrix}$$

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- We will have a lot to say about probability later in this course
- To reason about events with uncertain outcomes we introduce **random variables**
- Random variables corresponds to objects that we assign numbers to depending on the outcome of a random event
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- A continuous random variable might be the temperature at 12 noon tomorrow in Hyde park

Probability Distributions

- For discrete events, we can define a **probability mass function** $\mathbb{P}(X = x)$ that the random variable takes the value x
- This is a number between 0 and 1 with $\sum_{x \in \mathcal{X}} \mathbb{P}(X = x) = 1$
- For continuous events the probability of X taking a particular value x is zero
- Instead we consider the probability density $f_X(x)$ which we can define as

$$f_X(x) = \lim_{\delta x \rightarrow 0} \frac{\mathbb{P}(x < X \leq x + \delta x)}{\delta x}$$

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Expectations

- One of the most useful quantities in dealing with random numbers is to look at the average value, aka as the **expectation**, for some function $g(X)$ of the random variable

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- Expectations are linear operators

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Lesson

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- If you don't, you are going to struggle following any machine learning course, so grab a maths book and revise this
- We will revisit most of these topics to gain a better understanding, but you will be lost in the other machine learning modules if you are not familiar with everything I've covered
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